At least in the short term, AI and robotics are unlikely to completely eliminate entire industries. Jobs that require specialisation in a narrow range of routinised activities will be automated. But it will be much more difficult to replace humans with machines in less routine jobs that demand the simultaneous use of a wide range of skills, and that involve dealing with unforeseen scenarios. Take healthcare, for example. Many doctors focus almost exclusively on processing information: they absorb medical data, analyse it, and produce a diagnosis. Nurses, in contrast, also need good motor and emotional skills in order to give a painful injection, replace a bandage, or restrain a violent patient. Hence we will probably have an AI family doctor on our smartphone decades before we have a reliable nurse robot.[9] The human care industry – which takes care of the sick, the young and the elderly – is likely to remain a human bastion for a long time. Indeed, as people live longer and have fewer children, care of the elderly will probably be one of the fastest-growing sectors in the human labour market.

Alongside care, creativity too poses particularly difficult hurdles for automation. We don't need humans to sell us music any more – we can download it directly from the iTunes store – but the composers, musicians, singers and DJs are still flesh and blood. We rely on their creativity not just to produce completely new music, but also to choose among a mind-boggling range of available possibilities.

Nevertheless, in the long run no job will remain absolutely safe from automation. Even artists should be put on notice. In the modern world art is usually associated with human emotions. We tend to think that artists are channelling internal psychological forces, and that the whole purpose of art is to connect us with our emotions or to inspire in us some new feeling. Consequently, when we come to evaluate art, we tend to judge it by its emotional impact on the audience. Yet if art is defined by human emotions, what might happen once external algorithms are able to understand and manipulate human emotions better than Shakespeare, Frida Kahlo or Beyoncé?

After all, emotions are not some mystical phenomenon – they are the result of a biochemical process. Hence, in the not too distant future a machine-learning algorithm could analyse the biometric data streaming from sensors on and inside your body, determine your personality type and your changing moods, and calculate the emotional impact that a particular song – even a particular musical key – is likely to have on you.[10]

Of all forms of art, music is probably the most susceptible to Big Data analysis, because both inputs and outputs lend themselves to precise mathematical depiction. The inputs are the mathematical patterns of sound waves, and the outputs are the electrochemical patterns of neural storms. Within a few decades, an algorithm that goes over millions of musical